POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Fall

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the microcomputer Architecture. 5
   b) Compare the Harvard and Von- Neumann architecture. 5
   c) Draw and explain the timing diagram of MVI M,05H. 5
2. a) Write an AIP in 8085 microprocessors to add two-32 bit numbers stored in memory locations starting from 2000H and 3000H respectively. 8
   b) Define T-state. Draw the labelled timing diagram for the instruction XRA B. 7
3. a) Explain the functional block of 8237-DMA controller with its pin and internal register. 7
   b) Explain memory interfacing? How do you Interface two PROM and one RAM with 4kb and 2 kb memory respectively with 8085 microprocessor 8
4. a) What is the use of 8259 PIC? Draw and explain the functional block diagram of 8259 PIC. 8
   b) Explain the addressing modes of 8086 with suitable example? 7
5. a) What do you mean by identifiers and statements? Explain the following directives: 8
   TITLE, DOSSEG, MODEL, CODE, DATA, ENDP
   b) write a program in 8086 to display "MICROPROCESSOR IS EASY TO LEARN" in lower case. 7
6. a) Differentiate between synchronous and asynchronous serial data communication and explain the uses of BISYNC protocol in communication. 8
   b) What is Interrupt Vector Table (IVT)? Draw the IVT for 8086 microprocessor and explain different types of 8086 interrupts with respect to interrupt vector table. 7
7. Write short notes on: (Any two)
   a) Microprocessor vs. Micro controller
   b) Difference between absolute and partial decoding.
   c) RS-232C standard

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